Reg. No. | | | | | | | | | | |

[18MSCH3H02]

THIRD SEMESTER M.Sc. UNIVERSITY EXAMINATION, NOVEMBER - 2019

ORGANIC CHEMISTRY - III

Time:03 Hours **Maximum Marks:70**

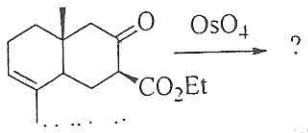
Instructions:
1. Question **1** is compulsory, carries **10** marks.
2. Questions **2 to 7** carry **15** marks each. Answer any Four.

1. Answer ANY FIVE of the following: **5x2=10**

a. Define asymmetric synthesis and asymmetric induction.

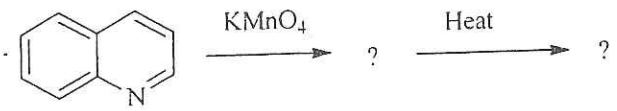
b. What is functional group interconversion (FGI)? Give an example.

c. Predict the product and mechanism for the following reaction.



d. Explain an intramolecular Diels-Alder reaction with an example.

e. Write the product in the following reaction



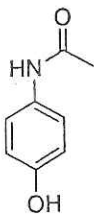
f. What is a hormone? Write the general structure of a steroid.

2a. Define Prelog's rule. Describe with suitable examples.

2b. Describe the Felkin-Anh-model. **10+5**

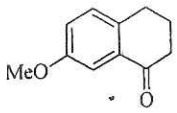
3a. Show the retrosynthesis of 3-nitrobenzaldehyde to get toluene as a starting material and discuss its synthesis.

3b. Show the disconnection approach for the target molecule paracetamol (**TM**) to get phenol as one of the starting material and explain its synthesis.



TM

3c. Show the appropriate disconnection and synthetic approach for tetralone (**TM**).



Tetralone (TM)

5+5+5

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4a. Explain the conversion mechanism from carbon-halogen to carbon hydride using trialkyltin hydrides.

4b. How does crown ether acts as a phase transfer catalyst? Explain with an example. **10+5**

5a. What is a [3,3] sigmatropic shift? Discuss the Cope rearrangement with the aspects of kinetics, stereochemistry, *oxy* and *aza* Cope rearrangements.

5b. Explain an electrocyclic reaction with an example? Outline its important features. **10+5**

6a. How to convert pyrimidine to pyrazole. Explain the mechanism.

6b. Discuss the oxidation reaction of indole with mechanism.

6c. Explain the Biginelli's synthesis of dihydropyrimidine with mechanism. **5+5+5**

7a. Discuss all properties of monosaccharides in detail.

7b. What are the differences between reducing and nonreducing sugars? **10+5**